

FedMed — One-page summary FedMed — One-page summary A short summary of what has been implemented on the feat/flower-monai-scaffold branch so far and how to view this page locally. Overview FedMed is a scaffold for privacy-preserving cross-silo federated learning for medical imaging. The scaffold is designed to be ported to Flower (federated orchestration), PyTorch + MONAI (3D U-Net segmentation), and a future TenSEAL homomorphic encryption integration. Implemented features (scaffold) Feature branch: feat/flower-monai-scaffold (local commits created) requirements.txt listing the primary dependencies (Flower, torch, MONAI, numpy, pytest) Basic Python package layout: fedmed/ with __init__.py Model package stub: fedmed/model/__init__.py and a placeholder for UNet3D Project directories created: model/, data/, fl/, encryption/, scripts/, and .github/workflows/ Documentation directory created: fedmed/docs/ (this file lives here) Encryption approach decided: add a clear encryption interface plus a TenSEAL integration stub/mock later (keeps CI and onboarding simple for now) Design decisions captured in the repository README describing the simplified NumPy-based reference implementations and porting notes How to open this page in your browser From Windows File Explorer: double-click fedmed/docs/one_pager.html. Or run this command from PowerShell or cmd: start "" "C:\Users\user\OneDrive\Desktop\Axlero Internship\FedMed\fedmed-federated-learning.worktrees\fedmed-federated-learning-privacy-ml\fedmed\docs\one_pager.html" Or open the file directly in most editors (VS Code: File → Open File...). Recommended next steps Implement UNet3D model prototype in fedmed/model/unet3d.py (MONAI-compatible API) Provide a Flower server and client template under fedmed/fl/ Add a small synthetic dataset and an example run script in scripts/ to demo one federated round locally Replace the TenSEAL stub with a working integration when native build support is available Created on branch feat/flower-monai-scaffold. Commits are local; push when ready.